

Sommation exercices - corrigé

1 Sommation 1

$$\begin{aligned}\sum_{i=k}^m \sum_{a=j}^n i &= \sum_{i=k}^m i \sum_{a=j}^n 1 \\&= \sum_{i=k}^m i \left(\sum_{a=0}^n 1 - \sum_{a=0}^{j-1} 1 \right) \\&= \sum_{i=k}^m i(n+1-j) \\&= (n+1-j) \sum_{i=k}^m i \\&= (n+1-j) \left(\sum_{i=0}^m i - \sum_{i=0}^{k-1} i \right) \\&= (n+1-j) \left(\frac{m(m+1)}{2} - \frac{k(k-1)}{2} \right) \\&= (n+1-j) \left(\frac{m^2+m}{2} - \frac{k^2-k}{2} \right) \\&= (n+1-j) \left(\frac{m^2+m-k^2+k}{2} \right) \\&= (n+1-j) \left(\frac{(m+k)(m-k+1)}{2} \right) \\&= \frac{(n+1-j)(m+k)(m-k+1)}{2}\end{aligned}$$

2 Sommmation 2

$$\begin{aligned}
 \sum_{i=1}^n (2i - 1) &= \sum_{i=1}^n 2i - \sum_{i=1}^n 1 \\
 &= 2 \sum_{i=1}^n i - \sum_{i=1}^n 1 \\
 &= 2 \left(\frac{n(n+1)}{2} \right) - n \\
 &= n^2 + n - n \\
 &= n^2
 \end{aligned}$$

3 Sommmation 3

$$\begin{aligned}
 \sum_{j=0}^{n-1} j2^j &= \sum_{j=0}^n (j2^j) - n2^n \\
 &= \frac{n2^{n+2} - (n+1)2^{n+1} + 2}{(1-2)^2} - n2^n \\
 &= n2^{n+2} - (n+1)2^{n+1} + 2 - n2^n \\
 &= n2^{n+2} - n2^{n+1} - 2^{n+1} + 2 - n2^n \\
 &= 2 + 2^n(4n - 2n - 2 - n) \\
 &= (n-2)2^n + 2
 \end{aligned}$$

4 Sommmation 4

$$\begin{aligned}
 \sum_{i=0}^{\beta} (i^2) - \sum_{j=0}^{\beta} (\beta - j) &= \sum_{i=0}^{\beta} (i^2) - \left(\sum_{j=0}^{\beta} \beta - \sum_{j=0}^{\beta} j \right) \\
 &= \left(\frac{\beta(\beta+1)}{2} \right)^2 - (\beta(\beta+1) - \frac{\beta(\beta+1)}{2}) \\
 &= \left(\frac{\beta(\beta+1)}{2} \right)^2 - \frac{\beta(\beta+1)}{2} \\
 &= \frac{\beta(\beta+1)}{2} \left(\frac{\beta(\beta+1)}{2} - 1 \right) \\
 &= \frac{\beta(\beta+1)}{2} \left(\frac{\beta^2 + \beta - 2}{2} \right) \\
 &= \frac{\beta(\beta+1)(\beta+2)(\beta-1)}{4}
 \end{aligned}$$

5 Sommatation 5

$$\sum_{d=n+1}^n d^4 = 0$$

6 Sommatation 6

$$\begin{aligned}\sum_{\alpha=1}^k \left(\frac{3}{2}\right)^{\alpha} &= \frac{1 - \left(\frac{3}{2}\right)^{k+1}}{1 - \frac{3}{2}} \\ &= \frac{1 - \left(\frac{3}{2}\right)^{k+1}}{-\frac{1}{2}} \\ &= 3\left(\frac{3}{2}\right)^k - 2\end{aligned}$$

7 Sommatation 7

$$\begin{aligned}\sum_{i=0}^{\log_3 n} \left(\frac{2}{3}\right)^i &= \frac{1 - \left(\frac{2}{3}\right)^{\log_3 n + 1}}{1 - \frac{2}{3}} \\ &= \frac{1 - \left(\frac{2}{3}\right)^{\log_3 n + 1}}{\frac{1}{3}} \\ &= 3 - 2\left(\frac{2}{3}\right)^{\log_3 n}\end{aligned}$$

8 Sommatation 8

$$\begin{aligned}\sum_{i=0}^{\log_3 n} n\left(\frac{2}{3}\right)^i &= n\left(\frac{1 - \left(\frac{2}{3}\right)^{\log_3 n + 1}}{1 - \frac{2}{3}}\right) \\ &= n\left(\frac{1 - \left(\frac{2}{3}\right)^{\log_3 n + 1}}{\frac{1}{3}}\right) \\ &= \frac{n}{3} - 2n\left(\frac{2}{3}\right)^{\log_3 n} \\ &= \frac{n}{3} - 2n\left(n^{\log_3 \frac{2}{3}}\right) \\ &= \frac{n}{3} - 2n^{\log_3 2}\end{aligned}$$

9 Sommatation 9

$$\begin{aligned}\sum_{k=2}^p k &= \sum_{k=0}^p k - \sum_{k=0}^1 k \\ &= \frac{k(k+1)}{2} - 1 \\ &= \frac{k^2 + k - 2}{2} \\ &= \frac{(k+2)(k-1)}{2}\end{aligned}$$

10 Sommatation 10

$$\begin{aligned}\sum_{p=0}^n (\sum_{p=0}^n 5 + \sum_{p=0}^n 2) &= \sum_{p=0}^n (5(n+1) + 2(n+1)) \\ &= \sum_{p=0}^n (7n+7) \\ &= \sum_{p=0}^n 7n + \sum_{p=0}^n 7 \\ &= 7n(n+1) + 7(n+1) \\ &= (n+1)(7n+7) \\ &= 7(n+1)^2\end{aligned}$$

11 Sommatation 11

$$\begin{aligned}\sum_{\theta=0}^n (\theta - \sum_{\psi=0}^m 2\psi) &= \sum_{\theta=0}^n (\theta - 2(m+1)) \\ &= \sum_{\theta=0}^n \theta - \sum_{\theta=0}^n (2m+2) \\ &= \frac{n(n+1)}{2} - (2m(n+1) + 2n+2) \\ &= \frac{n^2 + n - 4mn - 2m - 2n - 2}{2} \\ &= \frac{n^2 - n - 4mn - 2m - 2}{2}\end{aligned}$$

12 Sommatation 12

$$\begin{aligned}
 \sum_{i=1}^{\log n} \log\left(\frac{n}{2^i}\right) &= \sum_{i=1}^{\log n} \log n - \sum_{i=1}^{\log n} \log 2^i \\
 &= \eta \sum_{i=1}^{\eta} 1 - \sum_{i=1}^{\eta} i \quad \text{Posons } \eta = \log n \\
 &= \eta^2 - \frac{\eta(\eta+1)}{2} \\
 &= \frac{\eta^2 - \eta}{2} \\
 &= \frac{\log^2 n - \log n}{2}
 \end{aligned}$$

13 Sommatation 13

$$\begin{aligned}
 \sum_{s=0}^{n-1} s &= \sum_{s=0}^n s - s \\
 &= \frac{s(s+1)}{2} - s \\
 &= \frac{s^2 - s}{2} \\
 &= \frac{s(s-1)}{2}
 \end{aligned}$$

14 Sommatation 14

$$\begin{aligned}
 \sum_{k=2\sigma}^m k &= \sum_{k=0}^m k - \sum_{k=0}^{2\sigma-1} k \\
 &= \frac{m(m+1)}{2} - \frac{2\sigma(2\sigma-1)}{2} \\
 &= \frac{m^2 + m - 4\sigma^2 + 2\sigma}{2} \\
 &= \frac{(m+2\sigma)(m-2\sigma+1)}{2}
 \end{aligned}$$